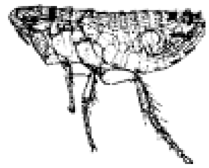
flea



NEWS 74 *June 2014*

http://esanetworks.org/

FLEA NEWS is a biannual newsletter about fleas (Siphonaptera). Recipients are urged to check any citations given here before including them in publications. Many of our sources are abstracting journals and current literature sources such as National Agricultural Library (NAL) Agricola, and National Library of Medicine (NLM) Medline, and citations have not necessarily been checked for accuracy or consistent formatting.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed in Agricola or Medline databases, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book reviews, Biography, Hypotheses and Literature Reviews, and Anecdotes. Send to:

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Assistant editor James R. Kucera

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Editorial

Dear Flea News Reader,

Observations of living things number in the billions, as recorded in museum collections and accumulated over centuries by thousands of scientists. Such records are a valuable resource for everyone.

Until recently, museum collections have been readily accessible only through a slow and laborious review. However, as Flea News 70 (June 2012, p. 4) mentioned, natural history museums are now building electronic databases.

Projects such as "Notes from Nature" (http://www.notesfromnature.org/?utm_source=Zooniverse%20Home&utm_medium=Web&utm_campaign=Homepage%20Catalogue#/archives) allow volunteers to transcribe and annotate handwritten and typed collection records into machine-readable (digitized) form. Researchers anywhere in the world could access such data through internet databases such as the Global Biodiversity Information Facility (GBIF) (http://www.gbif.org/), or iNaturalist.org (http://www.inaturalist.org/).

For fleas, these developments would allow us to answer more easily many interesting, heretofore difficult-to-address questions, such as where the distribution of flea species and subspecies is changing, what are the relationships of fleas to other species, especially their hosts, and how habitat types or abiotic characteristics, such as elevation and climate, affect fleas.

Generally, this type of research should be of immense significance for surmounting the "Taxonomic Impediment", which is the high necessity for, and low priority given, traditional taxonomy (Fontaine et al. 2012. PLoS ONE 7(5): e36881. doi: 10.1371/journal.pone.0036881). Better access to species information will assist in biotic conservation, and in understanding disease epidemiology, ecology, evolution, and environmental change, though the accessibility of the data raises new questions about preserving species from habitat loss and over-collecting.

Yours in fleas,
R.L. Bossard
Editor, Flea News

Additional information on the taxonomic impediment:

Cardoso, P., Erwin, T. L., Borges, P. A., & New, T. R. (2011). The seven impediments in invertebrate conservation and how to overcome them. Biological Conservation, 144(11), 2647-2655.

Dubois, A. (2010). Taxonomy in the century of extinctions: taxonomic gap, taxonomic impediment, taxonomic urgency. TAPROBANICA: The Journal of Asian Biodiversity, 2(1), 1-5.

Ebach, M. C., Valdecasas, A. G., & Wheeler, Q. D. (2011). Impediments to taxonomy and users of taxonomy: accessibility and impact evaluation. Cladistics, 27(5), 550-557.

United Nations Environment Programme Convention on Biological Diversity: http://www.cbd.int/gti/problem.shtml

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Announcements

New Doctoral Dissertations and Masters' Theses

Student: Andrade, Andréia Barbosa Navarro de
Title: Desenvolvimento de um sistema de liberação controlada de princípios ativos do óleo de *Azadirachta indica* A. Juss em matriz de poli (álcool vinílico) (PVA) para aplicações em veterinária

[Development of a controlled release system for active ingredients of oil of *Azadirachta indica* A. Juss in a matrix of poly (vinyl alcohol) (PVA) for veterinary applications]

Institution: Instituto de Pesquisas Energéticas e Nucleares, University of São Paulo, Brazil. 2013
Date: 2014

Sistemas de liberação controlada vêm sendo cada vez mais utilizados em diferentes áreas tendo em vista sua aplicabilidade que os tornam uma tecnologia capaz de atender as necessidades econômicas e ambientais uma vez que permitem a utilização de princípios ativos em quantidades precisas e apenas nos locais desejados. Dentre a diversidade de opções de matrizes utilizadas nesses sistemas o poli(álcool vinílico) tem sido bastante empregado e pode ser obtido pela técnica de ciclos térmicos. Azadiractina, um princípio ativo encontrado em óleo de semente de Nim tem sido descrita como repelente de diferentes tipos de insetos e apresenta grande aplicabilidade na agricultura. Com base nas propriedades desse composto, propôs-se o desenvolvimento de um sistema de liberação controlada para combate a pulgas *Ctenocephalides felis felis*. No presente trabalho, foram realizadas caracterizações físico-químicas das matrizes e dispositivos através de ensaios de fração gel, intumescimento, calorimetria exploratória diferencial e microscopia eletrônica de varredura e caracterizações biológicas quanto à citotoxicidade do óleo de Nim, matrizes e dispositivos e bioensaio para análise do comportamento das pulgas na presença do óleo de Nim. As caracterizações físico-químicas permitiram a escolha da matriz mais adequada para o desenvolvimento do dispositivo. As caracterizações biológicas demonstraram ausência de citotoxicidade do óleo de Nim, das matrizes e dos dispositivos e o bioensaio resultou em 40% de mortalidade das pulgas testadas. Na cinética de liberação, verificou-se que todos os agentes encapsulantes utilizados permitiram a liberação de princípios ativos do óleo de Nim.

[Controlled release systems are being increasingly used in different areas with a view of their application to technology meeting economic and environmental needs, as they allow the use of active ingredients in precise amounts and only at desired locations. Among the diversity of options of matrices used in these systems poly (vinyl alcohol) has been widely used and can be obtained by the technique of thermal cycles. Azadirachtin, an active principle found in Neem seed oil has been described as a repellent of different types of insects and has great applicability in agriculture. Based on the properties of the compound, it was proposed to develop a system for controlled release to combat the flea, *Ctenocephalides felis felis*. In this work, physicochemical characterizations of matrices and devices were made using tests of gel fraction, swelling, differential scanning calorimetry and scanning electron microscopy and biological characterizations for cytotoxicity of Neem oil, matrices and devices for analysis of the behavior of fleas in the presence of Neem oil by bioassay. The physicochemical characterizations allowed the choice of the most suitable matrix for the development of the device. The biological characterizations demonstrated no cytotoxicity of Neem oil, matrices and devices and the bioassay resulted in 40% mortality of fleas tested. On release kinetics, it was found that all encapsulating agents used allowed the release of active principles of Neem oil.]

Student: Clément, Christophe
Title: Le parasitisme des chats par les puces : étude de 490 cas présentés en consultation de dermatologie / parasitologie / mycologie à l'école vétérinaire de Nantes ONIRIS entre janvier 2008 et décembre 2012

[Parasitism of cats by fleas: a study of 490 cases presented in dermatologic consultation / parasitology / mycology at veterinary school of Nantes ONIRIS between January 2008 and December 2012]
Degree: Thèse d'exercice : Médecine vétérinaire
Institution: Université de Nantes, France.
Date: 2013.

L'auteur présente une étude sur l'infestation des chats par les puces à partir de l'analyse des dossiers cliniques des chats vus au service de dermatologie de l'Ecole Nationale Vétérinaire de Nantes en 2008 - 2009 - 2010. Il analyse des facteurs de risque (âge, sexe, race, ...), des caractéristiques de la dermatose (contagiosité, saisonnalité, prurit, ...) ainsi que les localisations et les lésions associées. Certaines tendances déjà évoquées dans des travaux similaires sont retrouvées (animaux d'intérieur sous exposés, lésions sur la ligne du dos principalement, ...) alors que d'autres apparaissent ici pour la première fois (prédisposition des femelles stérilisées à la DHPP, risque supérieur lors de la saison froide, ...).

[The author presents a study on the infestation of cats by fleas from the analysis of clinical records of cats seen at the dermatology department of the National Veterinary School of Nantes in 2008 - 2009 - 2010. It analyzes risk factors (age, sex, race, ...), the characteristics of the skin disease (contagiousness, seasonality, pruritus, ...) as well as locations and associated lesions. Some trends already mentioned in similar work are found (interior animals exposed, lesions principally on the back line, ...) while others appear here for the first time (sterilized female predisposition to DHPP [Distemper, Hepatitis, Parvo, and Parainfluenza], higher risk during the cold season, ...).]

Student: Fois, Francesco
Title: Phthiraptera Anoplura della Sardegna: analisi delle coinfestazioni ectoparassitiche rilevate su animali d'allevamento e selvatici.

[Phthiraptera Anoplura of Sardinia: analysis of ectoparasite coinfestation detected on farmed and wild animals.]
Degree: Doctoral thesis
Institution: Universita' degli Studi di Cagliari, Sardinia, Italy
Date: 23 May 2014

The studies on arthropods of medical and veterinary importance carried out in Sardinia until now not yet provide a clear and comprehensive overview on the taxa present, their distribution, phenology and the pathogen role. Phthiraptera Anoplura (= Siphunculata), which includes the so called "sucking lice" was almost never investigated in Sardinia. They are small insects, obligatory permanent ectoparasites, characterized by a high host specificity and a high ethological, physiological and anatomical specialization, including the wingless. They feed on epidermal tissues, and the entire life cycle of the Anoplura takes place on the host. Heavy louse infestation may cause to their hosts pruritus, alopecia, weakness, irritability, weight loss and anaemia. They are also potential vectors of pathogens that can cause severe disease in domestic and wild animals as well as humans. One of the most famous can be considered the epidemic typhus whose etiologic agent is *Rickettsia prowazeki* [*prowazekii*]. It is to be noted that the Anoplura have been little studied all around Italy and the few available published papers are occasional, sporadic and, apart from some recent works, already old. The aim of this PhD was to shed light on Phthiraptera Anoplura in Sardinia, taking into consideration the list of species present on the island, their distribution, phenology, and which mammals are parasitized. In this paper the Anoplura and other arthropod ectoparasites were collected on living and dead animals. The search for parasites was performed by examining the fur and skin of animals, concentrating on the areas of the body more vascularized and susceptible to localization of blood-sucking arthropods. So depending on the host species, was mainly look for the ectoparasites the regions of head, ears, neck, abdomen, the inguinal region and the perianal area. All arthropods found were fixed in 70% ethanol and later identified morphologically at the microscope. The collection of specimens in Sardinia provide information about twelve species of Phthiraptera Anoplura belonging to 6 families: Pediculidae (1 species), Phthiridae (1 species), Haematopinidae (4 species), Linognathidae (4 species), Hoplopleuridae (1 species), Echinophthiriidae (1 species). Were found 121 samples of arthropods with a total of 1294 specimens: 37 *Pediculus capitis* and 19 *Phthirus pubis* on man; 144 *Haematopinus apri* on wild boar; 1 *Haematopinus asini* on horse; 97 *Haematopinus suis* on pig and wild boar; 18 *Haematopinus tuberculatus* on buffalo; 787 *Linognathus africanus* on goat and sheep; 73 *Linognathus stenopsis* on goat and mouflon; 2 *Linognathus vituli* on cattle; 116 *Solenopotes burmeisteri* on Sardinian red deer. To add at the list, in literature two other species were cited for the island: *Schizophthirus pleurophaeus* found on the *Eliomys quercinus sardus* and *Echinophthirius horridus* found on *Monachus monachus* [Mediterranean monk seal]. Among other parasites (349 specimens) detected in coinfestation with Anoplura were identified 171 Acari Ixodida Amblyommidae [Ixodidae]: 73 *Rhipicephalus bursa* on goat, cattle, horse, sardinian red deer and mouflon; 50 *Rhipicephalus turanicus* on goat, sheep, pig, wild boar and mouflon; 8 *Haemaphysalis punctata* on goat and mouflon; 4 *Haemaphysalis sulcata* on mouflon; 31 *Dermacentor marginatus* on wild boar and mouflon; 5 *Hyalomma marginatum* on cattle and horse. On goat also were captured 7 fleas (Siphonaptera) belonging to *Ctenocephalides felis* (1 specimen); *Ctenocephalides canis* (1 specimen); *Pulex irritans* (5 specimens). On a horse was captured a specimen of *Hippobosca equina* (Diptera Ippoboscide [Hippoboscidae]). Were also found 170 Phthiraptera Ischnocera Bovicolidae: *Bovicola caprae* (143 specimens on goat); *Bovicola ovis* (23 specimens on sheep and mouflon); *Bovicola equi* (4 specimens on horse). These three species are at first founded in Sardinia. It has been reported the coinfestations between Anoplura and other parasites taxa in about the 80% of cases, on all the host species, except for man and buffalo. Here there are first reports for Sardinia for at least eight Phthiraptera species; among them *Linognathus africanus* was just recently reported on the Italian continental part, in Emilia Romagna, and is known in Europe only of Spain, Greece and Turkey. *Haematopinus tuberculatus was* also only recently reported of central-southern Italy. *Solenopotes burmeisteri*, was never previously officially reported for Italy but presumed. Besides the new species checklist a preliminary distribution maps for each species of Anoplura are provide. This study collect important new information about the species of ectoparasites that infest mammals in Sardinia, increasing meanwhile the knowledge about the same also in Italy and in the Mediterranean Basin. The results obtained also provide a starting point for faunistic, parasitological and epidemiological future works, for the protection and preservation of the regional livestock, wild animals and public health.

Student: González, Camila Abad
Title: Changes in mass of the preen gland in rock ptarmigans (*Lagopus muta*) in relation to sex, age and parasite burden 2007-2012.
Degree: Diss. M. Sc. Thesis.
Institution: Faculty of Life and Environmental Sciences, School of Engineering and Natural Sciences, University of Iceland, Reykjavík.
Date: January 2014
Advisors: Dr. Ólafur Karl Nielsen, Dr. Mariana Tamayo

The rock ptarmigan (*Lagopus muta*) is the main game-bird in Iceland and has a population that varies cyclically, usually peaking every 10 years. Given these population cycles and its cultural importance, the rock ptarmigan is studied throughout the year. Ptarmigan hunting is closely monitored and commerce of ptarmigans and their related products is forbidden since 2005 to assure sustainable harvest. The ptarmigan health project, which started in 2006, has focused on the general condition of the ptarmigans and how different health parameters relate to population changes during a population cycle. One of the parameters the project is studying is the preen gland. This gland is a holocrine organ exclusive to birds that produces preen oil, a secretion that birds spread through their plumage during preening. The preen oil has been proposed to offer protection against growth of feather degrading bacteria and fungi, as well as fighting ectoparasites. The present study aims to increase the understanding of the functions of the preen gland, specifically the changes in mass of the preen gland in rock ptarmigans in relation to sex, age, year and parasite burden, using data from ptarmigans collected in North-East Iceland from 2007-2012. The mean preen gland mass was significantly higher in males than in females, as well as in adult birds than in juvenile birds, but the effects of sex and age were no longer significant once corrected for body size. The year of collection had the greatest effect on the mass of the preen gland. Moreover, it was observed that the preen gland mass showed a significant negative relationship with ectoparasite richness and ectoparasite burden, as well as with the presence of chewing lice. This study provides evidence that the gland is a part of the ptarmigan outer defenses against ectoparasites.

"The only flea found on the ptarmigan is *Ceratophyllus garei*" (p. 16)

Student: Mfune, John Kazgeba
Title: Seasonal occurrence of fleas and other ectoparasites on small mammals at Waterberg Plateau, Namibia
Degree: Thesis, Master of Science in Biodiversity Management and Research
Institution: in collaboration with Humboldt-Universität zu Berlin
Date: 2014

Fleas and other ectoparasites infesting small mammals were studied from December 2005 to June 2006 at Waterberg Plateau Park, Namibia. The main aim of this study was to investigate seasonal occurrence of fleas and other ectoparasites on small mammals at selected sites at Waterberg Plateau Park. Small mammals were live-trapped, marked and released. Ectoparasites were collected (by brushing) from these small mammals before they were marked and released. Fleas, ticks and lice were stored in ethyl alcohol, whereas mites were stored in Oudeman's fluid before being processed and prepared for identification purposes. A total of one hundred and seventy nine (179) small mammals belonging to 11 rodent species (*Aethomys namaquensis*, *Aethomys chrysophilus*, *Dendromus melanotis*, *Graphiurus murinus*, *Gerbirullus paeba*, *Gerbirullus vallinus*, *Mus indutus*, *Mastomys* spp., Saccostomus campestris, *Tatera leucogaster* & *Thallomys nigricauda*) and 1 shrew (*Crocidura hirta*) were captured and examined. A total of one hundred and fourteen (114) fleas belonging to eight (8) species (*Listropsylla aricinae*, *Listropsylla dorippae*, *Pulex irritans*, *Xenopsylla brasiliensis*, *Xenopsylla cheopis*, *Xenopsylla nubica*, *Xenopsylla philoxera* & *Xenopsylla versuta*) and fifteen (15) ixodid ticks belonging to three (3) species (*Haemaphysalis elliptica*, *Hyaloma truncatum* [*Hyalomma truncatum*]& *Rhipicephalus neummani* [*Rhipicephalus neumanni*]) were recovered. Additionally, three lice (3) and four hundred & sixty nine (468[?]) mites were recovered but could not be identified. During the present study, a larva belonging to a tick species *R. neumanni* was collected for the first time from the wild. Again, a nymph of *R. neumanni* was recorded for only the second time in the wild in whole of Southern Africa. Generally, prevalence, intensity of infestation, and species diversity of ectoparasites varied among the 5 months, host species and between host sexes. Prevalence, intensity of infestation, and species diversity of ectoparasites was generally higher in December and lower in March; higher in *T. leucogaster* than *Mastomys* spp. & *G. paeba* and no ectoparasites were collected from *C. hirta*, *G. murinus*, *G. vallinus* and *S. campestris*. There was no significant difference in prevalence, intensity of infestation and species diversity of ectoparasites (except in prevalence of mites) between female and male hosts, fleas (prevalence: U=8.0, P=0.347; intensity: U=11.0, P=0.754; species diversity (H'): U=11.0, P=0.754); ticks (prevalence: U=9.0, P=0.465; intensity: U=11.0, P=0.735; species diversity (H'): U=7.5, P=0.296); mites (prevalence: U=2.0, P=0.028; intensity: U=9.0, P=0.465). However, fleas showed evidence of seasonality in the prevalence of infestation (H=5.58, df=4, P0.001).

Student: Parejo, S. Hervías
Title: Ecología e impacto de los mamíferos invasores en ecosistemas insulares: la isla de Corvo, Archipiélago de las Azores.
[Ecology and impact of invasive mammals on island ecosystems: Island of Corvo, Archipelago of Azores.]
Degree: Doctoral thesis
Institution: Ciencias de la Salud, Universidad de Murcia
Date: 5-Mar 2014

This research was focused on the direct and indirect impact of the three most common invasive mammals in insular ecosystems (house mouse *Mus domesticus*, black rat *Rattus rattus* and cat *Felis silvestris catus*) on Cory’s shearwater (*Calonectris diomedea borealis*). The work has been carried out on Corvo, the most remote island of the Azores Archipelago and the last to be discovered and colonized. Consequently, the introduction of invasive mammals occurred more recently (approximately 1600). Therefore, Corvo supports a high diversity of seabird species and the largest population of Cory's shearwater. Although colonies of other seabird species (e.g. Madeiran storm petrel *Oceanodroma castro* or little shearwater *Puffinus assimilis*) were numerous in a recent past, today their populations have strongly declined mainly due to predation by introduced mammals. The impact of these mammals on Cory's shearwater breeding success is unknown, however, this information is required to develop needed conservation measures. For each of the three invasive mammal species targeted in this study, evidence of their individual negative effects on seabird populations can be found within the scientific literature. However, the purpose of this research was to evaluate the overall impact of all coexisting invasive mammals on Corvo Island on the breeding success of Cory's shearwaters. To achieve this, different ecological aspects of these mammals were studied as well as their relationships. Combining data on trophic behavior with activity around the Cory's shearwater colonies could be observed two interactions established between cats and rats, which influenced the Cory's shearwater breeding success. Although only evidence for lethal relationships (predator-prey) has been demo[n]strated in seabirds, this research also addressed the indirect impact of predators influencing the ectoparasite intensity of Cory's shearwater. The four scientific studies that encompassed this research showed that cats are principally responsible for the low breeding success of Cory's shearwaters on Corvo Island. Among rodent species, only rats prey upon Cory's shearwater nests; however mice could act as predators in the absence of invasive mammals occupying a higher trophic level. Even if Cory's shearwaters were intensely preyed upon by cats during the breeding season, the availability of rats could limit cat predation of Cory's shearwater chicks. Both predators of Cory's shearwater (cats and rats) can compete for the same prey (Cory's shearwaters); however rats seem to have a little impact on Cory's shearwaters. This is probably due to the low abundance of rats and a high level of predation by cats as rats are their most important food source. Therefore, the predator behavior of rats could be moderated by cats. Lastly, the presence of predators in the vicinity of the nests increased Cory's shearwater ectoparasite intensity. This indirect impact of predators and ectoparasites on Cory's shearwaters deserved special attention because it affects body condition, nest survival, and, hence, the health of individuals. The information included in this research about the ecology of invasive mammals in insular ecosystems, is important for developing measures than can contribute to increase breeding success. Such increases can help address threats to the adult population (for example, bycatch in fishing nets) and contribute to the survival of Cory's shearwaters, thus avoiding irreversible damage as that suffered by neighboring species.

(extended extract) p. 36: "This examination identified two species of phthiraptera (*Austromenopon echinatum* and *Halipeurus abnormis*), one species of siphonaptera (*Xenopsylla gratiosa*), and two species of ticks (*Ixodes ricinus* and *Haemaphysalis punctata*). The highest intensity was recorded for *H. abnormis*. No blood parasites were detected and three species of nematodes (*Seuratia shipleyi*, *Contracaecum rudolphii* and *Thominx contorta*), one species of cestode (*Tetrabothrius minor*) and one trematode (*Cardiocephalus physalis*) were found."

Student: Suomalainen, Marjo
Title: Molecular Factors Affecting the Activity and Substrate Selectivity of the Pla Protease of *Yersinia pestis*

Institution: University of Helsinki, Finland, Faculty of Biological and Environmental Sciences, Department of Biosciences, Division of General Microbiology.
Date: 2014-04-04

Omptins are a family of conserved, integral outer membrane proteases and widely distributed within Gram-negative bacterial species. The family offers a good example of the evolution and the adaptation of a protein to novel functions and to differing pathogenic bacterial life-styles. This work investigates three different omptins: Pla of *Yersinia pestis*, PgtE of *Salmonella enterica* and OmpT of *Escherichia coli*. The omptin proteases differ in substrate specificity and need lipopolysaccharide (LPS) for activity. My thesis work addressed two main questions in omptin function: what is the molecular basis of the dissimilar substrate selectivity in the structurally very similar omptins; and what are the structural features in LPS that affect omptin activity. I studied the LPS dependency of omptins by expressing the proteins in bacterial cells that differ in LPS structure and by reconstituting purified, detergent-solubilized omptin protein with characterized, purified LPS molecules. *Y. pestis* alters its LPS structure in response to change of temperature from 20°C to 37°C, which reflects the transfer from a flea to a mammalian host. I found that the activity of Pla in cells from 20°C was very low, whereas cells from 37°C expressed high activity. I reconstituted detergent-purified His6-Pla protein with various model LPS structures and with LPSs of *Y. pestis* grown at different temperatures. Adding *Y. pestis* LPS from 37°C to the nonfunctional Pla protein induced high proteolytic activity, whereas 20°C-LPS gave very low activity, indicating that the activity of Pla is controlled by LPS. Similarly, I found that the activity of PgtE was high with rough LPS and low with smooth LPS; the difference mimics the LPS of intracellular (rough) and extracellular (smooth) *S. enterica*. Thus, in both bacterial species the omptin activity is controlled by the LPS type that the bacteria express during infection in mammals. I further studied the fine structure of *Y. pestis* LPS that affects Pla activity. This was done by reconstituting Pla activity with various structurally characterized *Y. pestis* and *E. coli* LPSs. I found that lower levels of lipid A acylation and phosphate substitution by aminoarabinose, are important for Pla activity, these features are characteristic for *Y. pestis* LPS from 37°C. A common and conserved feature in omptin structure is the presence of LPS-binding motif in protein barrel. Disrupting of the lipid A-binding motifs in PgtE and Pla abolished their proteolytic activity, emphasizing the importance of the LPS binding site for omptin activity. Omptins have a highly spatically [spatially] conserved active center and catalytic domains but express functional heterogeneity. The omptin transmembrane barrel contains five surface-exposed loops that show slightly higher sequence variation than the transmembrane protein regions. To study the effect of loop structures in omptin proteolytic specificity, I changed OmpT of *E. coli* to a Pla-like enzyme by a stepwise substitution of the loop areas. The proteins were characterized by their ability to activate the human protease precursor plasminogen (Plg) to the active serine protease plasmin and to inactivate the main plasmin inhibitor, α2-antiplasmin (α2AP); both functions are important for bacterial virulence. Pla cleaves very efficiently both substrates, whereas OmpT is only poorly active with them. I showed that OmpT could be converted into a Pla-like enzyme by cumulative substitutions at the loop areas, especially the loops L3-L5 were important. The successful conversion of OmpT towards Pla indicates that the loop structures are critical for omptin activity by

allowing correct recognition of the polypeptide substrate. More detailed substitution analysis was taken to identify the catalytic residues in Pla. My thesis demonstrates that the omptin proteolytic activity depends on two things: their specific interaction with LPS and the structure of their surface-exposed loops. The thesis offers an example of omptins extensive evolvability and of how they adapt to the lifestyle of their host bacterium.

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Meeting announcements

The Society for Vector Ecology (SOVE), 45th Annual Conference will meet on Sunday September 28, 2014 at 2:00 PM PDT-to-Thursday October 2, 2014 at 12:00 PM PDT, at the Menger Hotel, 204 Alamo Plaza, San Antonio, TX 78205.

The European SOVE Conference, October 13—17, 2014 will meet at the Makedonia Palace Hotel in Thessaloniki, Greece. Contact: eva.veronesi@pirbright.ac.uk http://www.sove.org/European%20SOVE%20folder/homeesove.html

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The Fifth International Forum for Sustainable Management of Disease Vectors will be held on November 2-6, 2014 in Qingdao, Shandong, China. www.chinavbc.cn/forum/

The International Symposium on Ectoparasites of Pets, the Livestock Insect Workers Conference, and the American Association of Veterinary Parasitologists will meet together in Boston next summer, holding their joint conference July 11-14, 2015. Plan to attend and present your research on fleas, ticks, lice, and other ectoparasites, including their health effects on pets, poultry, livestock, wildlife, and other animals. Deadlines and additional information will be posted at AAVP.org; to be added to the mailing list, send your e-mail address to NHinkle@uga.edu and you will receive updates.

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Research in Progress

Dr. Robert E. Lewis is writing a comprehensive treatise on the Fleas of North America.

Fleas on Bears in Wyoming, U.S.A.

Cam Lay reports that several fleas happened to be collected in February 2014 from a female black bear, *Ursus americanus* (Pallas), who had emerged early from hibernation in very poor condition on Rockefeller Parkway, administered by Grand Teton National Park (Wyoming, U.S.A.). Two of the fleas were female *Chaetopsylla tuberculaticeps* (Bezzi, 1890). One question is whether cat fleas, often found on urban wildlife (Dryden et al. 2011), infest bears foraging around campsites.

Chaetopsylla tuberculaticeps (Bezzi) lives in the NW U.S., Canada, Alaska, north of the Brooks Range in Alaska (Haas et al. 2005) and SE Alaska (Haas et al. 1989), as well as northern Asia and Europe, and southern alpine Europe. Rogers and Rogers (1976) wrote, "Fleas identified as *Chaetopsylla setosa* were reported from wild black bears in southern British Colombia [Columbia] (Rothschild 1906, Hopkins and Rothschild 1956) and Montana (Hubbard 1947) and from wild grizzly bears from British Columbia (Jellison and Good 1942, Ewing and Fox 1943, Holland 1949). A larger species of flea, *Chaetopsylla (Arctopsylla) tuberculaticeps ursi*, was reported from wild grizzly bears in southern Alberta (Rothschild 1902, Hopkins and Rothschild 1956), southern British Columbia (Holland 1949), and Alaska (Rausch 1961). The same subspecies was found on wild black bears in southcentral Alaska (Jellison and Kohls 1939) and in Montana (Hubbard 1947). In Montana, Jonkel and Cowan (1971) found it on 4 of 153 live-trapped black bears, and Worley et al. (1976) found *Chaetopsylla* spp. on 1 of 3 wild grizzly bears. Another subspecies, *Chaetopsylla tuberculaticeps tuberculaticeps*, was reported from brown bears from Norway, Russia and the Italian Alps (Hopkins and Rothschild 1956). The close taxonomic relationship between *C. t. tuberculaticeps* of Eurasia and *C. t. ursi* of North America is shown by the fact that fleas with characteristics intermediate between the two subspecies were collected from a wild brown bear (*Ursus arctos yesoensis*) on the island of Hokkaido, Japan (Sakaguti 1960). The specimens from Japan tentatively were classified as *C. t. tuberculaticeps* (Sakaguti op. cit.)..." (Rogers and Rogers 1976).

Haas et al. (2012) noted that, for many Alaskan mammals including black bears, "… their fleas are totally unknown. Some hosts might have just one or two accidental fleas, but such records are nonetheless of interest. These potential host mammals are the following: singing vole, snowshoe hare, pygmy shrew, dusky shrew, tundra shrew, Canadian lynx, red fox, American black bear, wolverine, American marten, ermine, least weasel, American mink. Collecting fleas from carnivores can require years in the field instead of months compared with small, abundant, easily trapped and handled rodents." Probably because of these hardships, the we understand the life cycles of Vermipsyllidae poorly, and no immature stages have been described (Elbel 1991).

Rogers and Rogers (1976) note that "Life cycles of fleas (*Chaetopsylla*) that infest bears have yet to be documented but probably are tied to the denning habits of their hosts. Fleas typically leave their host as adults to lay eggs in bedding material. There the eggs hatch, and the young develop and wait for a suitable host. For fleas to build up a high population on a given host, the host must return to beds where fleas have bred. Perhaps one reason that few black bears carry heavy populations of fleas is that the black bear tends to use a different den each winter and a different bed each night during the summer (unpublished data). However, a large population of fleas (*Orchopeas caedens*) was found in the recently abandoned den of a young bear in Minnesota (unpublished data). Presumably if certain fleas reach the den of a bear, they can build large populations during the denning period which is longer than six months in many cases." (Rogers and Rogers 1976). Squirrel fleas (*Orchopeas caedens*) do not parasitize bears as their primary host, but fleas occasionally thrive on secondary hosts, such as cat fleas (*Ctenocephalides felis*) infesting dairy calves and other domestic animals (Rust and Dryden 1997, Araujo et al. 1998).

There have been only a handful of observations on bear fleas in Montana in the last few decades. I hope that these new collections of bear fleas will be published, as an updated note would be of interest to entomologists, especially considering the on-going, rapid changes in our natural environment. I thank Cam Lay and Jim Kucera.

– R. L. Bossard, Editor

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GLENN E. HAAS, JAMES R. KUCERA, STEPHEN O. MACDONALD, and JOSEPH A. COOK. 2012. First flea (Siphonaptera) records for Kanuti National Wildlife Refuge, Central Alaska. J. ENTOMOL. SOC. BRIT. COLUMBIA 109: 6-11.

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Rust, M. K., & Dryden, M. W. (1997). The biology, ecology, and management of the cat flea. Annual Review of Entomology, 42(1), 451-473.

Featured research

Acosta-Gutiérrez, Roxana. 2014. Biodiversidad de Siphonaptera en México. Revista Mexicana de Biodiversidad del suplemento. Supl. 85: S345-S352, DOI: 10.7550/rmb.35267

Hernández-Camacho, N., S. Vergara-Pineda, R. Acosta-Gutiérrez and R. W. Jones. 2014. Nuevos registros de pulgas de tlacuaches *Didelphis virginiana* (Kerr 1792) en Querétaro, México. Therya, 5(1): 347-354. DOI: 10.12933/therya-14-187
[New records of fleas on Virginia opossums *Didelphis virginiana* (Kerr 1792) in Queretaro, Mexico]

The opossum *Didelphis virginiana* (Kerr 1792) is one of the eight species of marsupials that live in the northern region of America and is widely distributed in Mexico. There is a large literature in North America of studies on this species, including its ectoparasites. But in Mexico studies are scarce, and so we decided to investigate the opossum ectoparasite species richness in a suburban landscape in Querétaro, México.

Methods: During January to May of 2010, we used 12 tomahawk traps to capture opossums. Each animal was immobilized with tiletilamina [Tiletamine, an anesthetic], and during recumbence we examined the opossums for ectoparasites. Fleas were collected in vials with 70% alcohol, then dehydrated in 100% alcohol, and clarified with lactic acid for further identification.

Results: We found four flea species on the trapped opossums (n = 26), 14 males, 12 females. The species were *Ctenocephalides felis*, *Euhoplopsyllus glacialis affinnis* [*affinis*], *Polygenis martinezbaezi* and *Polygenis* sp. Two of these species [*Euhoplopsyllus glacialis affinis* and *Polygenis martinezbaezi*] are considered as new records for this marsupial host in Mexico, although these species have been recorded previously in this species in the United States and in other marsupial species in Mexico.

Discussion: These fleas are new records for the Mexican opossum parasite fauna. This study, although initially didactic, reveals relevant information about the flea species richness on opossums in fragmented habitats in a suburban landscape in Querétaro. Because fleas have potential as disease and parasite vectors, this new information may be helpful in assessing their effects on human and wildlife health.

Beaucournu J.-C. & Laudisoit A., 2014. Une nouvelle espèce malgache de puce du genre *Tsaractenus* (Siphonaptera, Ceratophyllidae, Leptopsyllinae). Bulletin de la Société entomologique de France, sous presse.
[A new Malagasy flea species of the genus *Tsaractenus*]

Beaucournu Lorvelec AD & O., 2014. Mise à jour taxonomique et répartition des puces du genre *Ctenophthalmus* (Siphonaptera, Ctenophthalmidae) dans la région paléarctique occidentale. Annales de la Société entomologique de France, sous presse.
[Current taxonomy and distribution of fleas of the genus *Ctenophthalmus* (Siphonaptera, Ctenophthalmidae) in the Western Palearctic region]

Beaucournu J.-C. & Goodman S., 2014. *Paractenopsyllus exspectatus* n. sp. (Siphonaptera, Ceratophyllidae, Leptopsyllinae) nouveau membre de ce genre endémique de Madgascar. Bulletin de la Société entomologique de France, soumis.
[*Paractenopsyllus exspectatus* n. sp. (Siphonaptera, Ceratophyllidae, Leptopsyllinae) new member of this endemic Madgascar genus]

Durden, L A, and J-C Beaucournu. 2014. Three new species of fleas belonging to the genus *Macrostylophora* from the three-striped ground squirrel, *Lariscus insignis*, in Java. Medical And Veterinary Entomology 2014 May 9. doi: 10.1111/mve.12059. [Epub ahead of print]

Three new species of fleas belonging to the genus *Macrostylophora* (Siphonaptera, Ceratophyllidae) are described from the three-striped ground squirrel, *Lariscus insignis*, from Tjibodas, West Java (Jawa Barat), Indonesia at an elevation of 1500 m. *Macrostylophora larisci* sp. n. is described from three male specimens, *Macrostylophora debilitata* sp. n. is described from one male and *Macrostylophora wilsoni* sp. n. is described from one female. Non-genital morphological characters of the female specimen, including ctenidial spine shapes and lengths, show that it is not the corresponding female for either *M. larisci* sp. n. or *M. debilitata* sp. n. It is unusual for three different species of congeneric fleas to parasitize the same host species in the same geographical location. These three new species represent the first known records of *Macrostylophora* from Java and they could be enzootic vectors between rodents of flea-borne zoonotic pathogens such as *Rickettsia typhi* and *Yersinia pestis*, both of which are established on Java. A list is provided of the 43 known species and 12 subspecies of *Macrostylophora* together with their known geographical distributions and hosts. A map depicting the distributions of known Indonesian (and Bornean) species of *Macrostylophora* is also included.

Bossard, Robert L. 2014. Flea (Siphonaptera) species richness in the Great Basin Desert and island biogeography theory. Journal of Vector Ecology 39 (1): 164-167.

Numbers of flea (Siphonaptera) species (flea species richness) on individual mammals should be higher on large mammals, mammals with dense populations, and mammals with large geographic ranges, if mammals are islands for fleas. I tested the first two predictions with regressions of H. J. Egoscue's trapping data on flea species richness collected from individual mammals against mammal size and population density from the literature. Mammal size and population density did not correlate with flea species richness. Mammal geographic range did, in earlier studies. The intermediate-sized (31 g), moderately dense (0.004 individuals/m^2) *Peromyscus truei* (Shufeldt) had the highest richness with eight flea species on one individual. Overall, island biogeography theory does not describe the distribution of flea species on mammals in the Great Basin Desert, based on H. J. Egoscue's collections. Alternatively, epidemiological or metapopulation theories may explain flea species richness.

Burrows, Malcolm and Gregory Sutton. 2013. Interacting Gears Synchronize Propulsive Leg Movements in a Jumping Insect. Science 13 September Vol. 341 no. 6151 pp. 1254-1256. DOI: 10.1126/science.1240284

Gears are found rarely in animals and have never been reported to intermesh and rotate functionally like mechanical gears. We now demonstrate functional gears in the ballistic jumping movements of the flightless planthopper insect *Issus*. The nymphs, but not adults, have a row of cuticular gear (cog) teeth around the curved medial surfaces of their two hindleg trochantera [*sic*, seems to be used as a plural form of trochanter. The usual form is trochanters.]. The gear teeth on one trochanter engaged with and sequentially moved past those on the other trochanter during the preparatory cocking and the propulsive phases of jumping. Close registration between the gears ensured that both hindlegs moved at the same angular velocities to propel the body without yaw rotation. At the final molt to adulthood, this synchronization mechanism is jettisoned.

Darvishi, Mohammad Mehdi, et al. 2014. A new flea from Iran. Asian Pacific Journal of Tropical Disease 4.2 (2014): 85-87.

Fleas are obligatory ectoparasites of humans and animals. These tiny insects are hematophagous and they can transmit a wide varity of disease agents to humans and domesticated animals. Indeed, this pest causes a considerable economic damages and health dangers particularly in tropical and subtropical. During an investigation on ectoparasites of five *Mus muscuuls* [*musculus*] in Semnan province, Iran, 15 fleas (8 males and 7 females) were collected. The extracted fleas mounted using clearing, dehydrating, mounting process and preserved with Canada balsam. After precise study, all of examined specimens were recognized as *Leptopsylla aethiopicus aethiopicus* using available systematic keys. This is the first report of this genus and species in Iran.

Grimaldi, David, and M. Andrew Johnston. 2014. The long-tongued Cretaceous scorpionfly *Parapolycentropus* Grimaldi and Rasnitsyn (Mecoptera: Pseudopolycentropodidae): New data and interpretations. American Museum Novitates 3793 (2014): 1-24.

The genus *Parapolycentropus*, originally described for two species in 99 myo Burmese amber, is unique among Mecoptera for its long, thin proboscis and possession of just the mesothoracic pair of wings. A new series of 19 specimens with excellent preservation allows description and redescription of virtually all morphological details.

Male terminalia are very similar to those of the Holarctic Recent family of "snow fleas," the Boreidae. Thoracic sclerites are highly convergent with nematocerous Diptera in the expansion of the mesothorax and great reduction of the pro-and metathoraces. The metathoracic wing vestige appears to be just the tegula; axial sclerites are lost. Details of the pretarsal claws are described; in *P. paraburmiticus* Grimaldi and Rasnitsyn the outer claw of the meso- and metathoracic pretarsi is elongate and the inner claw reduced. The proboscis is comprised not of a labial tube and "pseudolabellum" (contra Ren et al, 2009), but is mostly maxillary in origin, with the outer valves probably being galeae and the central, serrated stylet probably the hypopharynx. Abdominal sternites are greatly reduced (more so in females), suggesting that the abdomen was distensible, a feature that is common in some fluid-feeding insects. The proboscis, claw, and sternite modifications indicate that *Parapolycentropus* fed on the hemolymph of small insects, not the blood of vertebrates.

Hastriter, Michael W, and Sarah E Bush. 2014. Description of *Medwayella independencia* (Siphonaptera, Stivaliidae), a new species of flea from Mindanao Island, the Philippines and their phoretic mites, and miscellaneous flea records from the Malay Archipelago. Zookeys no. 408: 107-123.

Huang, Diying. 2014. *Tarwinia australis* (Siphonaptera: Tarwiniidae) from the Lower Cretaceous Koonwarra fossil bed: Morphological revision and analysis of its evolutionary relationship. Cretaceous Research (2014).Cretaceous Research Available online 9 May In Press.

Tarwinia australis, from the Lower Cretaceous Koonwarra fossil bed, Victoria, South Australia, is the first described Mesozoic flea. It was suggested to have a unique morphology that differs from all other known Mesozoic giant fleas by having a laterally-flattened body and peculiar tibial ctenidia. It represents an extinct family, Tarwiniidae, among the three major Mesozoic monster flea groups. Re-examination of the holotype reveals that this Southern Hemisphere ferocious bloodsucker bears different morphological details from those described previously. *Tarwinia australis* definitely bears elongate siphonate mouthparts and a relatively compact antenna with 15 flagellomeres. Its legs are slender, elongate, and armed with pseudocombs-like ctenidia on all tibiae. The abdomen is covered with posteriorly-directed setal rows, and with a posteriorly-located pygidium and exposed male genitalia. It differs from pseudopulicids mainly on the basis of characters of tibial ctenidia that probably indicate a very different host association.

Kamiya, T., O'Dwyer, K., Nakagawa, S. and Poulin, R. (2014) Host diversity drives parasite diversity: meta-analytical insights into patterns and causal mechanisms. Ecography, 37: 689–697. doi: 10.1111/j.1600-0587.2013.00571.x

Statistical correlations of biodiversity patterns across multiple trophic levels have received considerable attention in various types of interacting assemblages, forging a universal understanding of patterns and processes in free-living communities. Host–parasite interactions present an ideal model system for studying congruence of species richness among taxa as obligate parasites are strongly dependent upon the availability of their hosts for survival and reproduction while also having a tight coevolutionary relationship with their hosts. The present meta-analysis examined 38 case studies on the relationship between species richness of hosts and parasites, and is the first attempt to provide insights into the patterns and causal mechanisms of parasite biodiversity at the community level using meta-regression models. We tested the distinct role of resource (i.e. host) availability and evolutionary co-variation on the association between biodiversity of hosts and parasites, while spatial scale of studies was expected to influence the extent of this association. Our results demonstrate that species richness of parasites is tightly correlated with that of their hosts with a strong average effect size (r= 0.55) through both host availability and evolutionary co-variation. However, we found no effect of the spatial scale of studies, nor of any of the other predictor variables considered, on the correlation. Our findings highlight the tight ecological and evolutionary association between host and parasite species richness and reinforce the fact that host–parasite interactions provide an ideal system to explore congruence of biodiversity patterns across multiple trophic levels.

Maher, S.P., and R.M. Timm. 2014. Patterns of host and flea communities along an elevational gradient in Colorado. Canadian Journal Of Zoology 92, no. 5: 433-442.

Patterns in community composition across a landscape are the result of mechanistic responses and species interactions. Interactions between hosts and parasites have additional complexity because of the contingency of host presence and interactions among parasites. To assess the role of environmental changes within host and parasite communities, we surveyed small mammals and their fleas over a dynamic elevational gradient in the Front Range in Colorado, USA. Communities were characterized using several richness and diversity metrics and these were compared using a suite of frequentist and randomization approaches. We found that flea species richness was related to the number of host species based upon rarefaction, but no patterns in richness with elevation were evident. Values of diversity measures increased with elevation, representing that small-mammal and flea communities were more even upslope, yet turnover in composition was not related to examined variables. The results suggest there are strong local effects that drive these small-mammal and flea communities, although the breadth of flea species is tied to host availability.

Medvedev, S. G. 2014. The Palaearctic centers of taxonomic diversity of fleas (Siphonaptera). Entomological Review 94.3: 345-358.
2013. Entomologicheskoe obozrenie. T. XCII, N 4, P. 684-702.

Porta, D., Gonçalves, D. D., Gerônimo, E., Dias, E. H., Martins, L. de A., Ribeiro, L. V. P., Otutumi, L. K., Messa, V., & Gerbasi, A. V. (2014). Parasites in synanthropic rodents in municipality of the Northwest region of the State of Paraná, Brazil. African Journal of Microbiology Research, 8(16), 1684-1689.

Poulin, R. (2014) Parasite biodiversity revisited: frontiers and constraints. Int. J. Parasitol., http://dx.doi.org/10.1016/ j.ijpara.2014.02.003

Although parasites are widely touted as representing a large fraction of the Earth's total biodiversity, several questions remain about the magnitude of parasite diversity, our ability to discover it all and how it varies among host taxa or areas of the world. This review addresses four topical issues about parasite diversity. First, we cannot currently estimate how many parasite species there are on Earth with any accuracy, either in relative or absolute terms. Species discovery rates show no sign of slowing down and cryptic parasite species complicate matters further, rendering extrapolation methods useless. Further, expert opinion, which is also used as a means to estimate parasite diversity, is shown here to be prone to serious biases. Second, it seems likely that we may soon not have enough parasite taxonomists to keep up with the description of new species, as taxonomic expertise appears to be limited to a few individuals in the latter stages of their career. Third, we have made great strides toward explaining variation in parasite species richness among host species, by identifying basic host properties that are universal predictors of parasite richness, whatever the type of hosts or parasites. Fourth, in a geographical context, the main driver of variation in parasite species richness across different areas is simply local host species richness; as a consequence, patterns in the spatial variation of parasite species richness tend to match those already well-documented for free-living species. The real value of obtaining good estimates of global parasite diversity is questionable. Instead, our efforts should be focused on ensuring that we maintain sufficient taxonomic resources to keep up with species discovery, and apply what we know of the variation in parasite species richness among host species or across geographical areas to contribute to areas of concern in the ecology of health and in conservation biology.

Pucu E, M. Lareschi & SL Gardner. 2014. Bolivian Ectoparasites: A survey of the fleas of *Ctenomys* (Rodentia: Ctenomyidae). Comparative Parasitology 81 (1): 114-118.

We present the results of a multiyear survey of the fleas from ctenomyid rodents across many different habitats from throughout Bolivia. New species records for Bolivia include *Tiamastus palpalis* and *Ectinorus (Panallius) galeanus*. New records of fleas from *Ctenomys* in Bolivia include *Gephyropsylla klagesi*, *Sphinctopsylla inca*, and *Tetrapsyllus tristis*.

Ramezani-Awal-Riabi, Hamed, and Alireza Atarodi. 2014. A Survey on Fauna of Fleas (Order: Siphonaptera) of Cow and Sheep in Southern Khorasan-e-Razavi Province, Iran. Zahedan Journal of Research in Medical Sciences 15: 29-32.

Background: The aim of this study was to evaluate the abundance and fauna of the fleas that are the carrier pathogen in the South of Khorasan-e-Razavi province. Materials and Methods: In this cross-sectional study, for catch the fleas of cow and sheep were used 3 methods such as: light trap, sticky trap, and human bite. Results: A total number of 580 fleas from Gonabad and Bajestan cities were collected that all species were *Pulex iritans* [*irritans*]. Thirty five percent fleas were collected from cow and 65% from sheep. Conclusion: Considering that fleas are carrier pathogens and tend to feed on human blood, which is offered periodically at different seasons, barn spraying with appropriate insecticide was performed.

M. K. Rust, R. Vetter, I. Denholm, B. Blagburn, M. S. Williamson, S. Kopp, G. Coleman, J. Hostetler, W. Davis, N. Mencke, R. Rees, S. Foit, and K. Tetzner. 2014. Susceptibility of Cat Fleas (Siphonaptera: Pulicidae) to Fipronil and Imidacloprid using Adult and Larval Bioassays. Journal of Medical Entomology 51(3):638-643. doi 10.1603/ME13240

The monitoring of the susceptibility of fleas to insecticides has typically been conducted by exposing adults on treated surfaces. Other methods such as topical applications of insecticides to adults and larval bioassays on treated rearing media have been developed. Unfortunately, baseline responses of susceptible strains of cat flea, *Ctenocephalides felis* (Bouchè), except for imidacloprid, have not been determined for all on-animal therapies and new classes of chemistry now being used. However, the relationship between adult and larval bioassays of fleas has not been previously investigated. The adult and larval bioassays of fipronil and imidacloprid were compared for both field-collected isolates and laboratory strains. Adult topical bioassays of fipronil and imidacloprid to laboratory strains and field-collected isolates demonstrated that LD50s of fipronil and imidacloprid ranged from 0.11 to 0.40 nanograms per flea and 0.02 to 0.18 nanograms per flea, respectively. Resistance ratios for fipronil and imidacloprid ranged from 0.11 to 2.21. Based on the larval bioassay published for imidacloprid, a larval bioassay was established for fipronil and reported in this article. The ranges of the LC50s of fipronil and imidacloprid in the larval rearing media were 0.07–0.16 and 0.11–0.21 ppm, respectively. Resistance ratios for adult and larval bioassays ranged from 0.11 to 2.2 and 0.58 to 1.75, respectively. Both adult and larval bioassays provided similar patterns for fipronil and imidacloprid. Although the adult bioassays permitted a more precise dosage applied, the larval bioassays allowed for testing isolates without the need to maintain on synthetic or natural hosts.

SANCHEZ, J.P. & LARESCHI, M. (2014) New records of fleas (Siphonaptera: Ctenophthalmidae, Rhopalopsyllidae and Stephanocircidae) from Argentinean Patagonia with remarks on the morphology of *Agastopsylla boxi* and *Tiarapsylla argentina*. Revista Mexicana de Biodiversidad 85: 383-390, 2014. DOI: 10.7550/rmb.42071

A high diversity of fleas parasitizing sigmodontine rodents has been mentioned for Patagonia. Several of these fleas have been described having their type localities in the region, including several endemic taxa. For many species, however, the original descriptions are brief and there are no new morphological contributions. In the present study we report 8 species of fleas (Ctenophthalmidae, Rhopalopsyllidae and Stephanocircidae) parasitizing sigmodontine rodents from Argentinean Patagonia. Nineteen new parasite–host associations are reported and all studied fleas extend their known geographic range. Among them, *Tiarapsylla argentina* is mentioned for the first time for Patagonia; *Craneopsylla minerva*, *Sphinctopsylla ares*, *Polygenis (P.) platensis* and *Polygenis (P.) rimatus* are registered for the first time for Chubut, and *Agastopsylla boxi*, *Ectinorus (E.) ixanus* and *Ectinorus (E.) hapalus* for Santa Cruz, extending the southernmost limit of their geographical distribution. Also, for *A. boxi* and *T. argentina* we describe the morphology of the aedeagus, so far unknown. Results extend the morphological information of fleas and contribute to the knowledge of Patagonian biodiversity.

Sanchez, Juliana, and Marcela Lareschi. 2014. Two new species of *Neotyphloceras* (Siphonaptera: Ctenophthalmidae) from Argentinean Patagonia. Zootaxa 3784, no. 2: 159-170.

Two new species of *Neotyphloceras* Rothschild, parasites of sigmodontine rodents from Argentinean Patagonia, are described and illustrated: *N. crackensis* n. sp. and *N. pardinasi* n. sp. These species are compared with their morphologically closest relatives. Males are characterized by the shape of the upper lobe of the fixed process of clasper; the shape and chaetotaxy of the distal arm of sternum IX and by the shape of the crochet of the aedeagus; females by the contour of the distal margin of sternum VII. *Neotyphloceras pardinasi* n. sp. is reported from western Chubut Province, while *N. crackensis* n. sp. is known from the eastern regions of Chubut and Santa Cruz Provinces. With these reports, the geographical distribution of *Neotyphloceras* is extended to eastern Patagonia. A key to the species and subspecies of *Neotyphloceras* is provided.

Suntsov, V. V. (2014) Ecological aspects of the origin of *Yersinia pestis*, causative agent of the plague: Concept of intermediate environment. Contemporary Problems of Ecology 7.1 : 1-11.

Modern phylogenies of *Yersinia pestis* (Logh.), causative agent of the plague, constructed using molecular-genetic methods, do not receive a satisfactory functional and adaptive interpretation and are far from being ecologically valid. We have presented an ecological scenario of the origin of the causative agent of the plague through the transition of the initial pseudotuberculosis microbe *Yersinia pseudotuberculosis* O:1b to a free hostal ecological niche (and a new adaptive zone) under ultracontinental climatic conditions of the Late Pleistocene (Sartan time, 22000–15000 years ago) in southern Siberia and Central Asia. An intermediate environment, i.e., the “Mongolian marmot *Marmota sibirica*-flea *Oropsylla silatiewi*” parasitic system, where the process of adaptation development of the plague microbe took place, has been characterized. A scenario based on the major principles of the modern synthetic theory of evolution opens the way to an ecological-genetic synthesis of the problem of plague origin and is an appropriate model for developing a theory of molecular evolution of pathogenic (plaguelike) microorganisms.

Tree of Sex Consortium. (2014) Tree of Sex: A database of sexual systems. Sci. Data 1:140015 doi: 10.1038/sdata.2014.15.

The vast majority of eukaryotic organisms reproduce sexually, yet the nature of the sexual system and the mechanism of sex determination often vary remarkably, even among closely related species. Some species of animals and plants change sex across their lifespan, some contain hermaphrodites as well as males and females, some determine sex with highly differentiated chromosomes, while others determine sex according to their environment. Testing evolutionary hypotheses regarding the causes and consequences of this diversity requires interspecific data placed in a phylogenetic context. Such comparative studies have been hampered by the lack of accessible data listing sexual systems and sex determination mechanisms across the eukaryotic tree of life. Here, we describe a database developed to facilitate access to sexual system and sex chromosome information, with data on sexual systems from 11,038 plant, 705 fish, 173 amphibian, 593 non-avian reptilian, 195 avian, 479 mammalian, and 11,556 invertebrate species.

Vashchenok, V. S. 2014. Species composition, abundance, and annual cycles of fleas (Siphonaptera) on bank voles (*Clethrionomys glareolus*) in the western part of Vologda Province (Babaevo District). Entomological Review 94.3: 359-366.

A total of 383 fleas of 11 species were collected off 428 bank voles (*Clethrionomys glareolus*) captured near Babaevo (59°4′N, 35°8′E). Three of the species recorded (*Amphipsylla rossica*, *Doratopsylla dasycnema*, and *Palaeopsylla soricis*) were not typical of these rodents, infesting the latter occasionally from other animals, such as the common vole *Microtus arvalis* and shrews inhabiting adjacent or the same biotopes. *Peromyscopsylla bidentata*, infesting the bank vole in most part of its range, was recorded only as a single female. *Megabothris turbidus* was also very rare, being sporadically recorded from May to October; the examined territory probably lies at the northern boundary of its range. The most abundant species, *Ctenophthalmus uncinatus*, had two peaks of abundance: in April, when adults emerged from overwintered cocoons, and in July. Then their abundance decreased and the last occasional individuals were recorded till December. The year-round parasite *Amalaraeus penicilliger* was most abundant in winter. Other species were few in number. Adults of *Megabothris rectangulatus* were recorded from April to August with two peaks of abundance: in April, when they emerged from overwintered cocoons and in July, when the second generation emerged. The univoltine species *Peromyscopsylla silvatica* emerged in July–August and parasitized till September. *Rhadinopsylla integella* was the most abundant in October–December, but occasional specimens were recorded in January. The polyxenous species *Hystrichopsylla talpae* emerged in late July and occurred till September.

Fleas in Art and History

"Plague appeared first in recorded human history in 541 CE at Pelusium on the eastern edge of the Nile delta during the reign at Constantinople of the emperor Justinian (Procopius 551 CE). Its agent, the bacterium *Yersinia pestis*, shipped around the Mediterranean with rats, causing epidemic waves about every 15 years for 200 years. The 15-year cycles possibly reflect the diminishment of herd immunity in successive human generations. The historical consequences may well include the end of antiquity and the onset of the Middle Ages in Europe (Little 2007, Rosen 2007).

Plague then disappeared for 600 years from the historical record, reappearing as the Black Death in the 1340s in Central Asia and Crimea, reaching Italy just in time for the Renaissance. This second Pandemic lasted over 400 years, devastating perhaps half of Europe.

Plague reappeared as the third Pandemic in eastern Asia in the last part of the 19th century, spreading to much of the world. It reached western North America in 1899 at San Francisco and perhaps Los Angeles and Seattle (Adjemian et al. 2007). From these three sites waves of epidemic advance brought plague across western North America to the western edge of the short grass prairie within 40 years. The present distribution of endemic plague is, however, much more limited than its early epidemic scope."

– Foley, P. and J. Foley. 2010. Modeling susceptible infective recovered dynamics and plague persistence in California rodent–flea communities. Vector Borne Zoonotic Dis. 10: 59-67.

"Past history has indicated that fleas have largely been important for two diseases, plague and murine typhus, neither of which is known to have occurred in Alaska in historical time. Rausch and coworkers at the Arctic Health Research Center (pers. comm.) reported that they had isolated a gram negative bacterium from a varying hare near Anchorage that was asserted to be *Pasteurella* [*Yersinia*] *pestis*. Inasmuch as plague had not been found previously in Alaska, material was sent to the CDCA San Francisco Field Station for confirmation. According to Quan et al. (115), their studies showed the Alaskan bacteria to be closely related to *pestis* in many respects and, on the other hand, to bear some relationships to *Pasteurella* [*Yersinia*] *pseudotuberculosis*. The true identity of this organism has probably not yet been established. In many respects, I feel that a more logical explanation of the plague occurring in wild rodents of North America is the concept that plague spread with the Eurasian mammals into the New World during Pleistocene or earlier, rather than assuming it was introduced into the coastal areas of California at the end of the previous century and thence spread into the native animals. I feel that an investigation of this subject in Alaska warrants further study."

- Cluff E. Hopla. 1965. ALASKAN HEMATOPHAGOUS INSECTS, THEIR FEEDING HABITS AND POTENTIAL AS VECTORS OF PATHOGENIC ORGANISMS 1: THE SIPHONAPTERA OF ALASKA. AAL-TR-64-12 Vol I, ARCTIC AEROMEDICAL LABORATORY, AEROSPACE MEDICAL DIVISION, AIR FORCE SYSTEMS COMMAND, FORT WAINWRIGHT, ALASKA. OKLAHOMA UNIV. RESEARCH INST., NORMAN.

The German romantic E.T.A. Hoffmann (1776 – 1822) is not so well known today, but his fantastic tales inspired the *Nutcracker* ballet, *Coppelia*, and Offenbach's opera *Tales of Hoffmann*. The following extract is from his last book, *Meister Floh* (*Master Flea*).

– Editor

Master Flea
by E.T.A. Hoffmann

[a selection from Vol. 2, in Specimens of German Romance, selected and translated from various authors, in three volumes. Printed for Geo. B. Whittaker, Ave Maria Lane, by Thomas Davison, Whitefriars, London 1826. (Google Books)]

"He caught up the little box, which held the magic glass, and was on the point of dashing it against the floor with all his might, when suddenly Master Flea stood before him on the counterpane: he was in his microscopic form, and looked extremely graceful and handsome, in a glittering scale-breastplate, and highly-polished golden boots.

"Hold!" he cried; "hold, most respected friend; do not commit an absurdity. You would sooner annihilate a sun-moat than fling this little indestructible glass but a foot from you, while I am near. For the rest, though you were not aware of it, I was sitting, as usual, in the folds of your neckcloth, when you were at the honest bookbinder's, and therefore heard and saw all that passed. Just so I have been a party to your present edifying soliloquy, and have learnt several things from it. In the first place, you have shown the purity of your mind in all its glory, whence I infer that the decisive moment is fast approaching. Then too I have found that, in regard to the microscopic glass, I was in a great error. Believe me, my honoured friend, although I have not the pleasure to be a man, as you are, but only a flea—no simple one, indeed, but a graduate,--still I thoroughly understand human beings, amongst whom I so constantly live. Most frequently their actions appear to me very ridiculous, and even childish.--Do not take it ill, my friend; I speak it only as Master Flea. You are right; it would be a bad thing, and could not possibly lead to any good, if a man were able to spy thus, without ceremony, into the brains of his neighbours; still to the careless, lively, flea this quality of the microscopic glass is not in the least dangerous.

"Most honoured friend, and, as fortune soon will have it, most happy friend,--you know that my people are of a reckless, merry disposition, and one might say that they consisted of mere youthful springalds. With this I can, for my part, boast of a peculiar sort of wisdom, which in general is wanting to you children of men;--that is, I never do any thing out of season. To bite is the principal business of my life, but I always bite in the right time and right place; lay that to your heart, my worthy friend.

"I will now back from your hands, and faithfully preserve the gift, intended for you, and which neither that preparation of a man, called Swammerdamm, nor Leuwenhock, who wears himself out with petty envy, could possess. And now, my honoured Mr. Tyss, resign yourself to slumber. You will soon fall into a dreamy delirium, in which the great moment will reveal itself. At the right time I shall be with you again."

Master Flea disappeared, and the brilliance, which he had spread, faded away in the darkness of the chamber, the curtains of which were closely drawn."

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On the next page

"The Ghost of a Flea" (1819-1820), by William Blake (1757–1827, London, England), 21.5x16cm, tempera, gold leaf on mahogany panel.

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Flea research of historical interest

SPECIFIC ZOONOSES ACQUIRED BY MAN FROM NON-PRIMATES
Compiled by R. Traub & W.L. Jellison (1984).

(from an unpublished presentation at the University of Oklahoma by Robert Traub on evolution as a factor in the transmission of diseases transmitted to man from other animals, and based upon a hypothesis advanced by R. Traub & W.L. Jellison (1981), in "Rickettsiae and Rickettsial Diseases", edited by W. Burgdorfer & R.L. Anacker, Academic Press, New York, 650 pg., pp. 517-546).

[Fleas are involved in zoonoses so this tabulation is of interest, though out-of-date with the discovery of more than 30 new zoonoses since 1984 (Chronological List of Recently Emerged Infectious Agents/Diseases, http://www.hpa.org.uk/Topics/InfectiousDiseases/InfectionsAZ/EmergingInfections/GeneralInformation/emergEmergingInfectionsListofagents/).

Traub and Jellison hypothesized that zoonoses from taxa closely related to humans, such as Primates, tend to infect humans. Though biologists hypothesize that humans acquired HIV and a few other zoonoses from Primates, the overall data seem to refute the hypothesis: most zoonoses are from ungulates (especially Artiodactyla), a group less closely related to humans than Primates. The question of what encourages microbes to jump among host species we cannot answer, but environmental association and order diversity must be significant, too.

- R.L. Bossard, Editor]

A. LIST OF INFECTIONS [1]

DISEASE	ANIMALS [2]
I. BACTERIAL INFECTIONS	

(Leprosy is excluded but very recent work raises anew the possibility that in part this may be a zoonoses involving armadillos (Edentata) in Texas, or an anthroponosis.)

ANTHRAX	ARTIODACTYLA, PERISSODACTYLA
AVIAN TUBERCULOSIS	AVES
BACTEROIDES INFECTIONS	ARTIODACTYLA
BORDETELLA INFECTION	PERISSODACTYLA
BORRELIOSIS [3]	ARTIODACTYLA, PERISSODACTYLA, RODENTIA
BOTULISM	PERISSODACTYLA
BRUCELLOSIS (4 kinds)	ARTIODACTYLA (3), CARNIVORA (1)
CAMPYLOBACTERIOSIS	ARTIODACTYLA
CLOSTRIDIAL GANGRENE, ETC. (3 kinds)	ARTIODACTYLA (3), PERISSODACTYLA (1)
CORYNEBACTERIOSIS	ARTIODACTYLA, PERISSODACTYLA

ERYSIPELOID	ARTIODACTYLA, PERISSODACTYLA
FUSOBACTERIUM INFECTIONS	ARTIODACTYLA
GLANDERS	ARTIODACTYLA, PERISSODACTYLA
LEPTOSPIROSIS	ARTIODACTYLA, CARNIVORA, PERISSODACTYLA, RODENTIA
LISTERIOSIS	ARTIODACTYLA, PERISSODACTYLA, RODENTIA
LYME DISEASE(?) [3]	ARTIODACTYLA, RODENTIA
PASTEURELLA MULTOCIDA INFECTIONS	CARNIVORA
PASTEURELLOSIS	ARTIODACTYLA
PLAGUE	RODENTIA
RAT-BITE FEVERS	
1) STREPTOBACILLARY (HAVERHILL) TYPE	RODENTIA
2) SODOKU (SPIRILLUM INFECTION)	RODENTIA, CARNIVORA
SALMONELLOSIS	ARTIODACTYLA, AVES [4]
STREPTOCOCCAL INFECTIONS a, c	ARTIODACTYLA
TETANUS	ARTIODACTYLA, PERISSODACTYLA
TUBERCULOSIS (M. AVIUM) [5]	ARTIODACTYLA
TUBERCULOSIS (M. BOVIS)	ARTIODACTYLA
TUBERCULOSIS (?) (M. TUBERCULOSIS) c	CARNIVORA, ARTIODACTYLA
TULAREMIA	RODENTIA, LAGOMORPHA
VIBRIOSIS (s. str.)	ARTIODACTYLA
YERSINIOSIS a	AVES, CARNIVORA

II. CHLAMYDIAL INFECTIONS

ORNITHOSIS [6]	AVES, ARTIODACTYLA (?)

III. PROTOZOAN INFECTIONS

BABESIOSIS (2 kinds)	ARTIODACTYLA, RODENTIA
BALANTIDIASIS	ARTIODACTYLA
CRYPTOSPORIDIOSIS	ARTIODACTYLA
GIARDIASIS	RODENTIA, CARNIVORA
LEISHMANIASIS	
1) CUTANEOUS	CARNIVORA
2) MUCO-CUTANEOUS	RODENTIA
3) VISCERAL	CARNIVORA, RODENTIA
PNEUMOCYSTIS PNEUMONITIS (?) [7]	RODENTIA
SARCOCYSTIS INFECTIONS (2)	ARTIODACTYLA (2)
TOXOPLASMOSIS	CARNIVORA, ARTIODACTYLA, RODENTIA
TRYPANOSOMIASIS	
1) *T. CRUZI* (CHAGAS' DISEASE)	CARNIVORA, RODENTIA
2) *T. RHODESIENSE*	ARTIODACTYLA

IV. RICKETTSIAL INFECTIONS [8]

EPIDEMIC TYPHUS (FLYING SQUIRREL TYPE)	RODENTIA
MURINE TYPHUS	RODENTIA

Q FEVER	ARTIODACTYLA, RODENTIA
QUEENSLAND TICK TYPHUS b	RODENTIA
RICKETTSIALPOX b	RODENTIA
SCRUB TYPHUS (?) b	RODENTIA
SPOTTED FEVER b	RODENTIA, LAGOMORPHA (?)
TICK TYPHUS (OTHER TYPES) b	RODENTIA

V. VIRAL INFECTIONS

ARGENTINIAN HEMORRHAGIC FEVER	RODENTIA
BOLIVIAN HEMORRHAGIC FEVER	RODENTIA
BOVINE PSEUDOVESICULAR STOMATITIS	ARTIODACTYLA
BUNYAVIRAL GROUP-C INFECTIONS	RODENTIA
CALIFORNIA-GROUP INFECTIONS	ARTIODACTYLA, RODENTIA, LAGOMORPHA
CENTRAL EUROPEAN TICK-BORNE ENCEPHALITIS	ARTIODACTYLA, RODENTIA, INSECTIVORA, PERISSODACTYLA (?)
COLORADO TICK FEVER	RODENTIA
CONGO-CRIMEAN HEMORRAGIC FEVER b	ARTIODACTYLA, RODENTIA (?)
COWPOX c	ARTIODACTYLA
DUGBE VIRUS DISEASE	ARTIODACTYLA
ENCEPHALOMYOCARDITIS	RODENTIA
EQUINE ENCEPHALITIDES (3 kinds)	ARTIODACTYLA (?) (3), PERISSODACTYLA (?) (3), AVES (R?) (3)
FOOT AND MOUTH DISEASE	ARTIODACTYLA
HANTAAN VIRUS INFECTION	RODENTIA
HORSE POX	PERISSODACTYLA
INFLUENZA	
1) HUMAN(?) c (?)	ARTIODACTYLA
2) SWINE	ARTIODACTYLA
3) AVIAN (?) c	AVES
JAPANESE B ENCEPHALITIS	ARTIODACTYLA, AVES (?) (R?)
KEMEROVO VIRUS DISEASE	ARTIODACTYLA
KYASANUR FOREST DISEASE b	CHIROPTERA (?), RODENTIA
LASSA FEVER	RODENTIA(?)
(FOUR OTHER ARENA VIRUSES)	RODENTIA
LOUPING ILL a	ARTIODACTYLA, RODENTIA(?)
MUCAMBO FEVER	RODENTIA
MUROID VIRAL NEPHROPATHY	
1) FAR EASTERN	RODENTIA
2) SCANDINAVIA	RODENTIA
MURRAY VALLEY ENCEPHALITIS	CARNIVORA, AVES (R?), ARTIODACTYLA
NAIROBI SHEEP DISEASE	ARTIODACTYLA
NEWCASTLE DISEASE	AVES
OMSK HEMORRHAGIC FEVER (?) b	RODENTIA
ORF	ARTIODACTYLA

PARAINFLUENZA (SENDAI VIRUS)	RODENTIA, ARTIODACTYLA
POWASSAN ENCEPHALITIS b	RODENTIA
PSEUDOCOWPOX	ARTIODACTYLA
PSEUDORABIES	ARTIODACTYLA
QUARANFIL VIRUS DISEASE	AVES (?}
RABIES a	CARNIVORA, CHIROPTERA
RIFT VALLEY FEVER	ARTIODACTYLA
ROSS RIVER VIRUS	RODENTIA (?)
RUSSIAN SPRING-SUMMER ENCEPHALITIS b	ARTIODACTYLA, AVES (?), RODENTIA
SHEEP POX	ARTIODACTYLA
SINDBIS FEVER	AVES (?)
SWINE VESICULAR DISEASE (ENTEROVIRUS)	ARTIODACTYLA
TACARIBE VIRUS INFECTION	CHIROPTERA (?)
TAHYNA VIRUS INFECTION	LAGOMORPHA (?), ARTIODACTYLA, PERISSODACTYLA
TICK-BORNE ENCEPHALITIS	RODENTIA, INSECTIVORA (?)
VACCINIA	ARTIODACTYLA
VENEZUELAN EQUINE ENCEPHALOMYELITIS	RODENTIA (?), ARTIODACTYLA (?)
VESICULAR STOMATITIS	ARTIODACTYLA, PERISSODACTYLA
WADI MEDANI INFECTION	ARTIODACTYLA
WESSELBRON VIRUS DISEASE	ARTIODACTYLA
WEST NILE FEVER	AVES

VI. FUNGAL INFECTIONS

DERMATOPHYTOSES	CARNIVORA, PERISSODACTYLA, RODENTIA

VII. UNKNOWN ETIOLOGY

CAT-SCRATCH DISEASE [9]	CARNIVORA

B. THE ZOONOSES ARRANGED BY ORDER, ETC. OF HOST

I. PLANTS (NIL)
II. INSECTS (NIL)
III. OTHER INVERTEBRATES (NIL)
IV. FISH (NIL)
V. AMPHIBIA (NIL)
VI. REPTILIA (NIL)
VII. AVES
- 1. BACTERIAL INFECTIONS
  - a. AVIAN TUBERCULOSIS
  - b. SALMONELLOSIS a
  - c. YERSINIOSIS a
- 2. CHLAMYDIAL INFECTIONS
  - a. ORNITHOSIS [6]

3. VIRAL INFECTIONS
    a. EQUINE ENCEPHALITIDES (?) (3 kinds)
    b. INFLUENZA: AVIAN (?)
    c. JAPANESE B ENCEPHALITIS
    d. MURRAY VALLEY ENCEPHALITIS
    e. NEWCASTLE DISEASE
    f. QUARANFIL VIRUS DISEASE (?)
    g. RUSSIAN SPRING-SUMMER ENCEPHALITIS
    h. WEST NILE FEVER

VIII. MARSUPIALIA (NIL)

IX. INSECTIVORA
1. VIRAL INFECTIONS
    a. TICK-BORNE ENCEPHALITIS b

X. CHIROPTERA
1. VIRAL INFECTIONS
    a. KYASANUR FOREST DISEASE (?)
    b. RABIES a
    c. TACARIBE VIRUS INFECTIONS (?)

XI. LAGOMORPHA
1. BACTERIAL INFECTIONS
    a. TULAREMIA a
2. RICKETTSIAL INFECTIONS
    a. SPOTTED FEVER (?) b
3. VIRAL INFECTIONS
    a. CALIFORNIA-GROUP INFECTIONS I
    b. TAHYNA VIRUS INFECTION (?)

XII. CARNIVORA
1. BACTERIAL INFECTIONS
    a. BRUCELLOSIS (1 kind)
    b. LEPROSPIROSIS
    c. PASTEURELLA MULTOCIDA INFECTIONS
    d. RAT-BITE FEVER: SODOKU (SPIRILLUM INFECTION)
    e. TUBERCULOSIS (?) (*M. TUBERCULOSIS*) c
    f. YERSINIOSIS a
2. VIRAL INFECTIONS
    a. MURRAY VALLEY ENCEPHALITIS
    b. RABIES a
3. PROTOZOAN INFECTIONS
    a. CUTANEOUS LEISHMANIASIS
    b. GIARDIASIS
    c. TOXOPLASMOSIS
    d. TRYPANOSOMIASIS (*T. CRUZ*I) (CHAGAS' DISEASE)
    e. VISCERAL LEISHMANIASIS
4. FUNGAL INFECTIONS
    a. DERMATOPHYTOSES
5. UNKNOWN ETIOLOGY

a. CAT-SCRATCH DISEASE [9]

XIII. RODENTIA

1. BACTERIAL INFECTIONS
    a. BORRELIOSIS
    b. LEPTOSPIROSIS
    c. LISTERIOSIS
    d. LYME DISEASE (?) [10]
    e. PLAGUE
    f. RAT-BITE FEVER: SODOKU (SPIRILLUM INFECTION)
    g. RAT-BITE FEVER: STREPTOBACILLARY (HAVERHILL) TYPE
    h. TULAREMIA
2. RICKETTSIAL INFECTIONS
    a. EPIDEMIC TYPHUS (FLYING-SQUIRREL TYPE)
    b. MURINE TYPHUS
    c. Q FEVER
    d. QUEENSLAND TICK TYPHUS b
    e. RICKETTSIALPOX b
    f. SCRUB TYPHUS (?) b
    g. SPOTTED FEVER
    h. TICK TYPHUS (OTHER TYPES) b
3. PROTOZOAN INFECTIONS
    a. BABESIOSIS (1 kind)
    b. GIARDIASIS
    c. CUTANEOUS LEISHMANIASIS
    d. MUCO-CUTANEOUS LEISHMANIASIS
    e. PNEUMOCYSTIS PNEUMONITIS (?)
    f. TOXOPLASMOSIS
    g. TRYPANOSOMIASIS (*T. CRUZI*) (CHAGAS' DISEASE)
4. VIRAL INFECTIONS
    a. ARGENTINIAN HEMORRHAGIC FEVER
    b. BOLIVIAN HEMORRHAGIC FEVER
    c. BUNYAVIRAL GROUP-C INFECTIONS
    d. CALIFORNIA-GROUP INFECTIONS
    e. CENTRAL EUROPEAN TICK-BORNE ENCEPHALITIS
    f. COLORADO TICK FEVER
    g. ENCEPHALOMYOCARDITIS
    h. HANTAAN VIRUS INFECTION
    i. KYASANUR FOREST DISEASE (?) b
    j. LASSA FEVER
    k. (FOUR OTHER ARENA VIRUSES)
    l. LOUPING ILL a
    m. MUROID VIRAL NEPHROPATHY (FAR EASTERN)
    n. MUROID VIRAL NEPHROPATHY (SCANDINAVIAN)
    o. OMSK HEMORRHAGIC FEVER (?) b
    p. PARAINFLUENZA (SENDAI VIRUS)
    q. POWASSAN ENCEPHALITIS b

r. ROSS RIVER VIRUS (?)
s. RUSSIAN SPRING-SUMMER ENCEPHALITIS b
t. TICK-BORNE ENCEPHALITIS
u. VENEZUELAN EQUINE ENCEPHALOMYELITIS
5. FUNGAL INFECTIONS
a. DERMATOPHYTOSES
XIV. PERISSODACTYLA
1. BACTERIAL INFECTIONS
a. ANTHRAX
b. BORRELIOSIS
c. BOTULISM
d. CLOSTRIDIAL GANGRENE, ETC. (1)
e. CORYNEBACTERIOSIS
f. ERYSIPELOID
g. GLANDERS
h. LEPTOSPIROSIS
i. LISTERIOSIS
j. TETANUS
2. VIRAL INFECTIONS
a. CENTRAL EUROPEAN TICK-BORNE ENCEPHALITIS
b. EQUINE ENCEPHALITIDES (?) (3 kinds)
c. HORSE POX
d. VESICULAR STOMATITIS
3. FUNGAL INFECTIONS
a. DERMATOPHYTOSES
XV. ARTIODACTYLA
1. BACTERIAL INFECTIONS
a. ANTHRAX
b. BACTERIODES INFECTIONS
c. BORRELIOSIS VIRAL INFECTIONS
d. BRUCELLOSIS (3 kinds)
e. CAMPYLOBACTERIOSIS
f. CLOSTRIDIAL GANGRENE, ETC. (3)
g. CORYNEBACTERIOSIS
h. ERYSIPELOID
i. FUSOBACTERIUM INFECTIONS
j. GLANDERS
k. LEPTOSPIROSIS
l. LISTERIOSIS
m. LYME DISEASE(?) [10]
n. PASTEURELLOSIS
o. SALMONELLOSIS
p. STREPTOCOCCAL INFECTIONS a, c
q. TETANUS
r. TUBERCULOSIS (*M. BOVIS*)
s. TUBERCULOSIS (*M. AVIUM*)

t. TUBERCULOSIS(?) (*M. TUBERCULOSIS*) c
u. VIBRIOSIS (s. str.)
2. CHLAMYDIAL INFECTIONS
a. ORNITHOSIS (?) [6]
3. PROTOZOAN INFECTIONS
a. BABESIOSIS
b. BALANTIDIASIS
c. CRYPTOSPORIDIOSIS
d. SARCOCYSTIS INFECTIONS (2)
e. TOXOPLASMOSIS
f. TRYPANOSOMIASIS (*T. RHODESIENSE*)
4. RICKETTSIAL INFECTIONS
a. Q FEVER
5. VIRAL INFECTIONS
a. BOVINE PSEUDOVESICULAR STOMATITIS
b. CALIFORNIA-GROUP INFECTIONS
c. CENTRAL EUROPEAN TICK-BORNE ENCEPHALITIS
d. CONGO-CRIMEAN HEMORRHAGIC FEVER b
e. COWPOX c
f. DUGBE VIRUS DISEASE
g. EQUINE ENCEPHALITIDES (?) (3 kinds)
h. FOOT AND MOUTH DISEASE
i. INFLUENZA: HUMAN (?) c (?)
j. INFLUENZA: SWINE
k. JAPANESE B ENCEPHALITIS
l. KEMEROVO VIRUS DISEASE
m. LOUPING ILL a
n. NAIROBI SHEEP DISEASE
o. ORF
p. PARAINFLUENZA (SENDAI VIRUS)
q. PSEUDOCOWPOX
r. PSEUDORABIES
s. RIFT VALLEY FEVER
t. RUSSIAN SPRING-SUMMER ENCEPHALITIS
u. SHEEP POX
v. VACCINIA
w. VENEZUELAN EQUINE ENCEPHALITIS (?)
x. VESICULAR STOMATITIS
y. WADI MEDANI INFECTION
z. WESSELBRON VIRUS DISEASE

a = Several major sources.
b = Acarine vector may (also) be the reservoir.
c = Also passed in some degree from man to other animals.
R = Reservoir.

[Editors' notes:
1. Categories are often out-of-date, what with numerous limitations and additions to the literature. The list excludes multicellular pathogens, such as parasitic helminths (trematodes, cestodes, nematodes, and acanthocephalans).
2. Animal vectors and reservoirs.
3. Spirochaetal *Borrelia* bacteria cause Lyme Disease.
4. Salmonellosis is also vectored by REPTILIA and AMPHIBIA.
5. *Mycobacterium bovis* is in *M. tuberculosis* complex. *M. avis*, in *M. avium* complex.
6. Also known as psittacosis, or parrot fever.
7. Pathogen of PNEUMOCYSTIS PNEUMONITIS is now considered a fungus, not a protozoan.
8. *Rickettsiales* includes *Anaplasma*, *Ehlichia*, *Neorickettsia*, *Rickettsia*, and *Wolbachia*.
9. *Bartonella* bacteria are implicated in cat-scratch disease.
10. Lyme Disease reservoir is Rodentia.]

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Directory of Siphonapterists (Updated)

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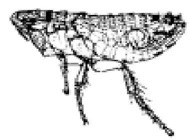

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